

Introduction to Electric Vehicles

Environmental as well as economical issues provide a compelling impetus to develop clean, efficient, and sustainable vehicles for urban transportation. Automobiles constitute an integral part of our everyday life, yet the exhaust emissions of conventional internal combustion (IC) engine vehicles are to blame for the major source of urban pollution that causes the greenhouse effect leading to global warming.¹ The dependence on oil as the sole source of energy for passenger vehicles has economical and political implications, and the crisis will inevitably become acute as the oil reserve of the world diminishes. The number of automobiles on our planet doubled to about a billion or so in the last 10 years. The increasing number of automobiles being introduced on the road every year is only adding to the pollution problem. There is also an economic factor inherent in the poor energy conversion efficiency of combustion engines. Although the number for alternative electric vehicles is not significantly higher when efficiency is evaluated on the basis of conversion from crude oil to traction effort at the wheels, it makes a difference. Emission due to power generation at localized plants is much easier to regulate than that emanating from IC engine vehicles (ICEV) that are individually maintained and scattered. People dwelling in cities are not exposed to power plant related emissions, because these are mostly located outside urban areas. Electric vehicles (EV) enabled by high-efficiency electric motors and controllers and powered by alternative energy sources provide the means for a clean, efficient, and environmentally friendly urban transportation system. Electric vehicles have no emission, having the potential to curb the pollution problem in an efficient way. Consequently, EVs are the only zero-emission vehicles possible.

Electric vehicles paved their way into public use as early as the middle of the 19th century, even before the introduction of gasoline-powered vehicles.² In the year 1900, 4200 automobiles were sold, out of which 40% were steam powered, 38% were electric powered, and 22% were gasoline powered. However, the invention of the starter motor, improvements in mass production technology of gas-powered vehicles, and inconvenience in battery charging led to the disappearance of the EV in the early 1900s. However, environmental issues and the unpleasant dependence on oil led to the resurgence of interest in EVs in the 1960s. Growth in the enabling technologies added to environmental and

economic concerns over the next several decades, increasing the demand for investing in research and development for EVs. Interest and research in EVs soared in the 1990s, with the major automobile manufacturers embarking on plans for introducing their own electric or hybrid electric vehicles. The trend increases today, with EVs serving as zero-emission vehicles, and hybrid electric vehicles already filling in for ultralow-emission vehicles.

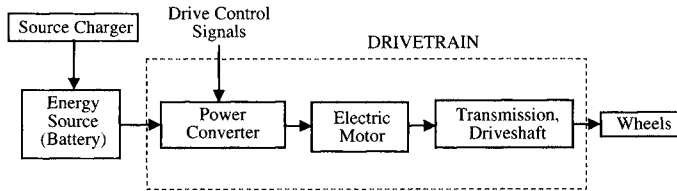


FIGURE 1.1 Top-level perspective of an EV system.

1.1 EV SYSTEM

An EV has the following two features:

1. The energy source is portable and chemical or electromechanical in nature.
2. Traction effort is supplied only by an electric motor.

Figure 1.1 shows an EV system driven by a portable energy source. The electromechanical energy conversion linkage system between the vehicle energy source and the wheels is the drivetrain of the vehicle. The drivetrain has electrical as well as mechanical components.

1.1.1 COMPONENTS OF AN EV

The primary components of an EV system are the motor, controller, power source, and transmission. The detailed structure of an EV system and the interaction among its various components are shown in Figure 1.2. Figure 1.2 also shows the choices available for each of the subsystem level components. Electrochemical batteries have been the traditional source of energy in EVs. Lead-acid batteries have been the primary choice, because of their well-developed technology and lower cost, although promising new battery technologies are being tested in many prototype vehicles. The batteries need a charger to restore the stored energy level once its available energy is near depletion due to usage. Alternative energy sources are also being developed for

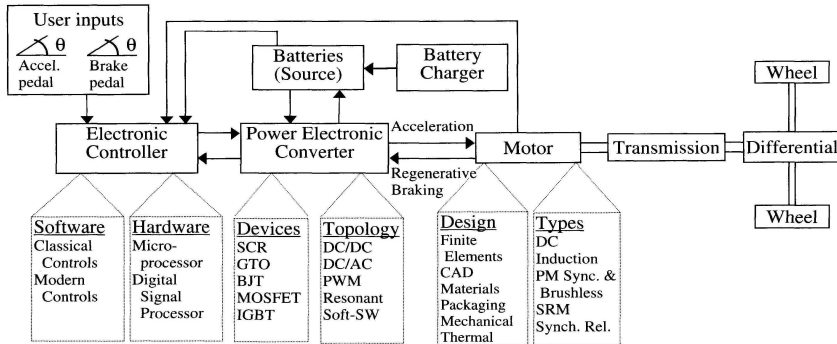


FIGURE 1.2 Major electrical components and choices for an EV system.

zero-emission vehicles. The limited range problem of battery-driven EVs prompted the search for alternative energy sources, such as fuel cells and flywheels. Prototypes have been developed with fuel cells, while production vehicles will emerge in the near future.

The majority of electric vehicles developed so far are based on DC machines, induction machines, or permanent magnet machines. The disadvantages of DC machines pushed EV developers to look into various types of AC machines. The maintenance-free, low-cost induction machines became an attractive alternative to many developers. However, high-speed operation of induction machines is only possible with a penalty in size and weight. Excellent performance together with high-power density features of permanent magnet machines make them an attractive solution for EV applications, although the cost of permanent magnets can become prohibitive. High-power density and a potentially low production cost of switched reluctance machines make them ideally suited for EV applications. However, the acoustic noise problem has so far been a deterrent for the use of switched reluctance machines in EVs. The electric motor design includes not only electromagnetic aspects of the machine but also thermal and mechanical considerations. The motor design tasks of today are supported by finite element studies and various computer-aided design tools, making the design process highly efficient.

The electric motor is driven by a power-electronics-based power-processing unit that converts the fixed DC voltage available from the source into a variable voltage, variable frequency source controlled to maintain the desired operating point of the vehicle. The power electronics circuit comprised of power semiconductor devices saw tremendous development over the past 3 decades. The enabling technology of power electronics is a key driving force in developing efficient and high-performance power-train units for EVs. High-power devices in

compact packaging are available today, enabling the development of lightweight and efficient power-processing units known as power electronic motor drives. Advances in power solid state devices and very large-scale integration (VLSI) technology are responsible for the development of efficient and compact power electronics circuits. The developments in high-speed digital signal processors or microprocessors enable complex control algorithm implementation with a high degree of accuracy. The controller includes algorithms for the motor drive in the inner loop as well as system-level control in the outer loop.

1.2 EV HISTORY

The history of EVs is interesting. It includes the insurgence of EVs following the discovery of electricity and the means of electromechanical energy conversion and later being overtaken by gasoline-powered vehicles. People digressed from the environmentally friendly mode of transportation due to lack of technology in the early years, but they are again focused on the correct track today.

1.2.1 THE EARLY YEARS

Prior to the 1830s, the means of transportation was only through steam power, because the laws of electromagnetic induction, and consequently, electric motors and generators, were yet to be discovered. Faraday demonstrated the principle of the electric motor as early as in 1820 through a wire rod carrying electric current and a magnet, but in 1831 he discovered the laws of electromagnetic induction that enabled the development and demonstration of the electric motors and generators essential for electric transportation. The history of EVs in those early years up to its peak period in the early 1900s is summarized below:

- Pre-830—Steam-powered transportation
- 1831—Faraday's law, and shortly thereafter, invention of DC motor
- 1834—Nonrechargeable battery-powered electric car used on a short track
- 1851—Nonrechargeable 19 mph electric car
- 1859—Development of lead storage battery
- 1874—Battery-powered carriage
- Early 1870s—Electricity produced by dynamo-generators
- 1885—Gasoline-powered tricycle car
- 1900—4200 automobiles sold:
 - 40% steam powered
 - 38% electric powered
 - 22% gasoline powered

The specifications of some of the early EVs are given below:

- 1897—French Krieger Co. EV: weight, 2230 lb; top speed, 15 mph; range, 50 mi/charge
- 1900—French B.G.S. Co. EV: top speed, 40 mph; range, 100 mi/charge
- 1912—34,000 EVs registered; EVs outnumber gas-powered vehicles 2-to-1
- 1915—Woods EV: top speed, 40 mph; range, 100 mi/charge
- 1915—Lansden EV: weight, 2460 lb, top speed, 93 mi/charge, capacity, 1 ton payload
- 1920s—EVs disappear, and ICEVs become predominant

The factors that led to the disappearance of EV after its short period of success were as follows:

1. Invention of starter motor in 1911 made gas vehicles easier to start.
2. Improvements in mass production of Henry T (gas-powered car) vehicles sold for \$260 in 1925, compared to \$850 in 1909. EVs were more expensive.
3. Rural areas had limited access to electricity to charge batteries, whereas gasoline could be sold in those areas.

1.2.2 1960s

Electric vehicles started to resurge in the 1960s, primarily due to environmental hazards being caused by the emissions of ICEVs. The major ICEV manufacturers, General Motors (GM) and Ford, became involved in EV research and development. General Motors started a \$15 million program that culminated in the vehicles called Electrovair and Electrovan. The components and specifications of two Electrovair vehicles (Electrovair I (1964) and Electrovair II (1966) by GM) are given below.

Systems and characteristics:

Motor—three-phase induction motor, 115 hp, 13,000 rev/m

Battery—silver-zinc (Ag-Zn), 512 V, 680 lb

Motor drive—DC-to-AC inverter using a silicon-controlled rectifier (SCR)

Top speed—80 mi/h

Range—40 to 80 miles

Acceleration—0–60 mi/h in 15.6 s

Vehicle weight—3400 lb

2

Vehicle Mechanics

The fundamentals of vehicle design are embedded in the basic mechanics of physics, particularly in Newton's second law of motion relating force and acceleration. Newton's second law states that *the acceleration of an object is proportional to the net force exerted on it*. The object accelerates when the net force is nonzero, where the term "net force" refers to the result of the forces acting on the object. In the vehicle system, several forces act on it, with the resultant or net force dictating the motion according to Newton's second law. A vehicle moves forward with the aid of the force delivered by the propulsion unit overcoming the resisting forces due to gravity, air, and tire resistance. The acceleration and speed of the vehicle depend on the torque and power available from the traction unit and the existing road and aerodynamic conditions. Acceleration also depends on the composite mass of the vehicle, including the propulsion unit, all mechanical and electrical components, and the batteries.

A vehicle is designed based on certain given specifications and requirements. Furthermore, the electric and hybrid vehicle system is large and complex, involving multidisciplinary knowledge. The key to designing such a large system is to divide and conquer. The system-level perspective helps in mastering the design skills for a complex system, where the broad requirements are first determined and then system components are designed with more focused guidelines. For example, first, the power and energy requirements from the propulsion unit are determined from a given set of vehicle cruising and acceleration specifications. The component-level design begins in the second stage, where the propulsion unit, the energy source, and other auxiliary units are specified and designed. In this stage, the electrical and mechanical engineers start designing the electric motor for electric vehicles (EVs) or the combination of electric motor and internal combustion (IC) engine for hybrid electric vehicles (HEVs). The power electronics engineers design the power conversion unit that links the energy source with the electric motor. The controls engineer works in conjunction with the power electronics engineer to develop the propulsion control system. The chemists and the chemical engineers have the primary responsibility of designing the energy source based on the energy requirement and guidelines of the vehicle manufacturer. Many of the component designs proceed in an

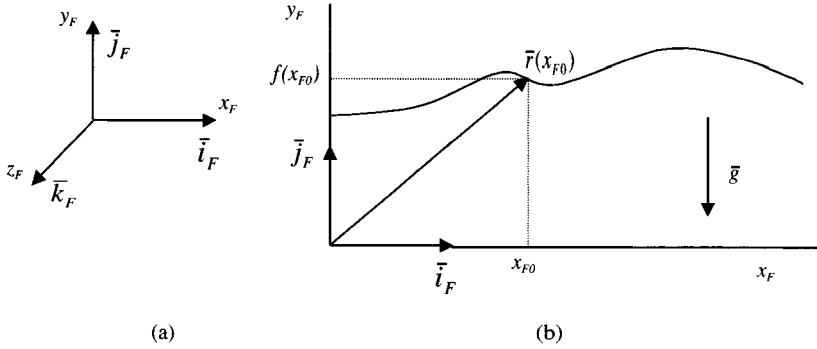


FIGURE 2.1 (a) Fixed coordinate system. (b) Roadway on the fixed coordinate system.

iterative manner, where various designers interact to ensure that the design goals are met.

Design is an iterative process, which starts with some known factors and other educated estimates or assumptions to be followed by scientific analysis in order to verify that the requirements are met. In this chapter, we will develop the tools for scientific analysis of vehicle mechanics based on Newton's second law of motion. After defining and describing a roadway, the vehicle kinetics issues will be addressed. The roadway and kinetics will be linked to establish the equation for the force required from the propulsion unit. The force from the propulsion unit, which can be an electric motor or an IC engine or a combination of the two, is known as *tractive force*, F_{TR} . Once the force requirement is defined, one would proceed to calculate the power and energy required for a vehicle under consideration. The emphasis of study in this book is on the broad design goals, such as finding the power and energy requirements and predicting the range for a given energy source, thereby maintaining a top-level perspective. The design details of a subsystem are beyond the scope of this book, and readers are referred to the literature of the respective areas for further details.

2.1 ROADWAY FUNDAMENTALS

A vehicle moves on a level road and also up and down the slope of a roadway. We can simplify our description of the roadway by considering a straight roadway. Furthermore, we will define a tangential coordinate system that moves along with the vehicle, with respect to a fixed two-dimensional system. The roadway description will be utilized to calculate the distance traversed by a vehicle along the roadway.

The fixed coordinate system is attached to the earth such that the force of gravity is perpendicular to the unit vector $\bar{i}_F$ and in the $x_F y_F$ plane. Let us

consider a straight roadway, i.e., the steering wheel is locked straight along the X_F direction. The roadway is then in the $x_F y_F$ plane of the fixed coordinate system (Figure 2.1).

The two-dimensional roadway can be described as $y_F = f(x_F)$. The *roadway position vector* $\bar{r}(x_F)$ between two points a and b along the horizontal direction is

$$\bar{r}(x_F) = x_F \bar{i}_F + f(x_F) \bar{j}_F \quad \text{for } a \leq x_F \leq b$$

The direction of motion and the distance traversed by the vehicle are easier to express in terms of the *tangent vector* of the roadway position vector, given as:

$$\bar{T}(x_F) = \frac{d\bar{r}}{dx_F} = \bar{i}_F + \frac{df}{dx_F} \bar{j}_F$$

The distance-norm of the tangent vector $\|\bar{T}(x_F)\|$ is

$$\|\bar{T}(x_F)\| = \sqrt{1 + \left[\frac{df}{dx_F} \right]^2}$$

The tangential roadway length s is the distance traversed along the roadway. Mathematically, s is the arc length of $y_F = f(x_F)$ over $a \leq x_F \leq b$. Therefore,

$$s = \int_a^b \|\bar{T}(x_F)\| dx_F$$

The *roadway percent grade* is the vertical rise per 100 horizontal distance of roadway, with both distances expressed in the same units. The angle β of the roadway associated with the slope or grade is the angle between the tangent vector and the horizontal axis X_F (Figure 2.2). If Δy is the vertical rise in meters, then

$$\% \text{ grade} = \frac{\Delta y}{100 \text{ m}} 100\% = \Delta y \%$$

The tangent of the slope angle is

$$\tan \beta = \frac{\Delta y}{100 \text{ m}}$$

The percent grade, or β , is greater than zero when the vehicle is on an upward slope and is less than zero when the vehicle is going downhill. The roadway percent grade can be described as a function of the roadway, as

$$\beta(x_F) = \tan^{-1} \left[\frac{df(x_F)}{dx_F} \right]$$

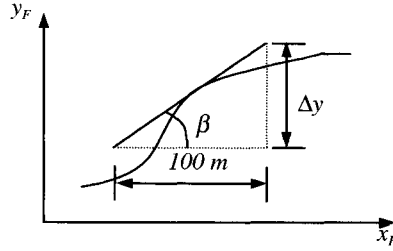


FIGURE 2.2 Grade of the roadway.

EXERCISE 2.1

A straight roadway has a profile in the $x_F y_F$ -plane given by $f(x_F) = 3.9 \sqrt{x_F}$ for $0 \leq x_F \leq 2$ miles (x_F and y_F are given in feet).

- (a) Plot the roadway.
- (b) Find $\beta(x_F)$.
- (c) Find the percent grade at $X_F=1$ mile.
- (d) Find the tangential road length between 0 and 2 miles.

Solution

- (b) $\tan^{-1} \frac{1.95}{\sqrt{x_F}}$
- (c) 2.68%
- (d) 10,580 ft

2.2 LAWS OF MOTION

Newton's second law of motion can be expressed in equation form as follows:

$$\sum_i \bar{F}_i = m\bar{a}$$

where

$$\sum_i \bar{F}_i$$

is the net force, m is the effective mass, and $\bar{a}$ is the acceleration. The law is applied to the vehicle by considering a number of objects located at several points of contact of the vehicle with the outside world on which the individual forces act. Examples of such points of contact are the front and rear wheels touching the roadway surface and vehicle, the frontal area that meets the force from air resistance, etc. We will simplify the problem by merging all of these points of contact into one location at the center of gravity (cg) of the EV and HEV, which is justified, because the extent of the object is immaterial. For all of

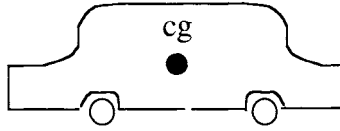


FIGURE 2.3 Center of gravity (cg) of a vehicle.

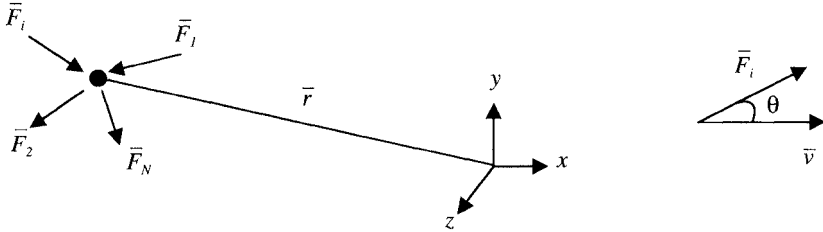


FIGURE 2.4 Forces on a particle.

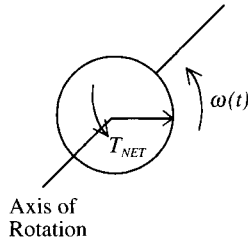


FIGURE 2.5 Rigid body rotation.

the force calculations to follow, we will consider the vehicle to be a particle mass located at the cg of the vehicle. The cg can be considered to be within the vehicle, as shown in [Figure 2.3](#).

Particle motion is described by particle velocity and acceleration characteristics. For the position vector $\bar{r}$ for the particle mass on which several forces are working, as shown in [Figure 2.4](#), the velocity v and acceleration a are

$$\bar{v} = \frac{d\bar{r}}{dt} \quad \text{and} \quad \bar{a} = \frac{d\bar{v}}{dt}$$

The power input to the particle for the i th force is

$$P_i = \bar{F}_i \cdot \bar{v} = |\bar{F}_i| |\bar{v}| \cos \theta$$

where θ is the angle between $\bar{F}_i$ and the resultant velocity v .

For a rigid body rotating about a fixed axis, the equivalent terms relating motion and power are torque, angular velocity, and angular acceleration. Let there be i independent torques acting on a rigid body, causing it to rotate about an axis of rotation, as shown in [Figure 2.5](#). If J (unit: $\text{kg}\cdot\text{m}^2$) is the polar moment

of inertia of the rigid body, then the rotational form of Newton's second law of motion is

$$\sum_i T_i = T_{NET} = J\alpha$$

where

ω = angular speed (rad/s)

$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$ = angular acceleration (rad/s²)

The power input for the i th torque is $P_i = T_i\omega$.

2.3 VEHICLE KINETICS

The tangential direction of forward motion of a vehicle changes with the slope of the roadway. To simplify the equations, a tangential coordinate system is defined below, so that the forces acting on the vehicle can be defined through a one-directional equation. Let $\bar{u}_T(x_F)$ be the *unit tangent vector* in the fixed coordinate system pointing in the direction of increasing X_F . Therefore,

$$\bar{u}_T(x_F) = \frac{\bar{T}_F(x_F)}{\|\bar{T}_F(x_F)\|} = \frac{\bar{i}_F + \frac{df}{dx_F} \bar{j}_F}{\sqrt{1 + \left[\frac{df}{dx_F}\right]^2}}$$

The tangential coordinate system shown in [Figure 2.6](#) has the same origin as the fixed coordinate system. The z -direction unit vector is the same as that in the fixed coordinate system, but the x - and y -direction vectors constantly change with the slope of the roadway.

Newton's second law of motion can now be applied to the cg of EV in the tangential coordinate system as

$$\sum \bar{F}_T = m\bar{a}_T = m \frac{d\bar{v}_T}{dt}$$

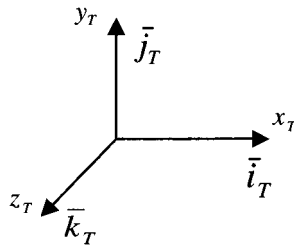


FIGURE 2.6 Tangential coordinate system and the unit tangent vector on a roadway.

where m is the effective vehicle mass. The components of the coordinate system are:

Component tangent to the road:

$$\sum \bar{F}_{xT} = m \frac{d\bar{v}_{xT}}{dt}$$

Component normal to the road:

$$\sum \bar{F}_{yT} = m \frac{d\bar{v}_{yT}}{dt}$$

And, because motion is assumed to be confined to the $x_T y_T$ plane:

$$\sum \bar{F}_{zT} = m \frac{d\bar{v}_{zT}}{dt} = 0$$

The vehicle tangential velocity is v_{xT} . The gravitational force in the normal direction is balanced by the road reaction force and, hence, there will be no motion in the y_T normal direction. In other words, the tire always remains in contact with the road. Therefore, the normal velocity $v_{yT}=0$. Vehicle motion has been assumed confined to the $x_F y_F$ or $x_T y_T$ plane and, hence, there is no force or velocity acting in the z -direction. These justified simplifications allow us to use a one-directional analysis for vehicle propulsion in the x_T -direction. It is shown in the following that the vehicle tractive force and the opposing forces are all in the x_T -direction. Hence, the vector “-” will not be used in the symbols for simplicity.

The propulsion unit of the vehicle exerts a tractive force, F_{TR} , to propel the vehicle forward at a desired velocity. The tractive force must overcome the opposing forces, which are summed together and labeled as the *road load force*, F_{RL} . The road load force consists of the gravitational force, rolling resistance of the tires, and the aerodynamic drag force. The road load force is as follows:

$$F_{RL} = F_{gxT} + F_{roll} + F_{AD} \quad (2.1)$$

where x_T is the tangential direction along the roadway. The forces acting on the vehicle are shown in [Figure 2.7](#).

The gravitational force depends on the slope of the roadway. The force is positive when climbing a grade and is negative when descending a downgrade roadway. The gravitational force to be overcome by the vehicle moving forward is

$$F_{gxT} = mg \sin \beta \quad (2.2)$$

where m is the total mass of the vehicle, g is the gravitational acceleration constant, and β is the grade angle with respect to the horizon.

The rolling resistance is produced by the hysteresis of the tire at the contact surface with the roadway. In a stationary tire, the normal force to the road balances the distributed weight borne by the wheel at the point of contact along the vertical line beneath the axle. When the tire rolls, the centroid of the vertical forces on the wheel moves forward from beneath the axle toward the direction of

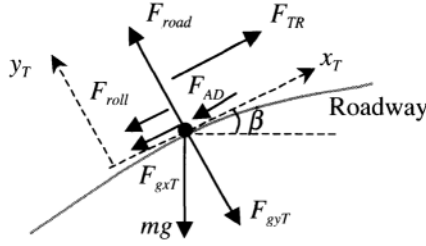


FIGURE 2.7 Forces acting on a vehicle.

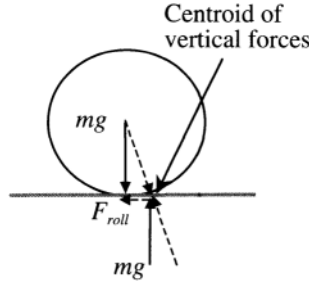


FIGURE 2.8 Rolling resistance force of wheels.

motion of the vehicle, as shown in Figure 2.8. The weight on the wheel and the road normal force are misaligned due to the hysteresis of the tire. They form a couple that exerts a retarding torque on the wheel. The *rolling resistance force*, F_{roll} , is the force due to the couple, which opposes the motion of the wheel. The force F_{roll} is tangential to the roadway and always assists in braking or retarding the motion of the vehicle. The tractive force F_{TR} must overcome this force F_{roll} along with the gravitational force and the aerodynamic drag force. Rolling resistance can be minimized by keeping the tires as inflated as possible as well as by reducing the hysteresis. The ratio of the retarding force due to rolling resistance and the vertical load on the wheel is known as the *coefficient of rolling resistance*, C_0 . The rolling resistance force is given by the following:

$$F_{roll} = \begin{cases} \text{sgn}[v_{xT}]mg(C_0 + C_1v_{xT}^2) & \text{if } v_{xT} \neq 0 \\ (F_{TR} - F_{gxT}) & \text{if } v_{xT} = 0 \text{ and } |F_{TR} - F_{gxT}| \leq C_0 mg \\ \text{sgn}[F_{TR} - F_{gxT}](C_0 mg) & \text{if } v_{xT} = 0 \text{ and } |F_{TR} - F_{gxT}| > C_0 mg \end{cases} \quad (2.3)$$

Typically, $0.004 < C_0 < 0.02$ (unitless), and $C_1 < C_0$ (s^2/m^2). $C_0 mg$ is the maximum rolling resistance at standstill. The $\text{sgn}[v_{xT}]$ is the signum function given as

$$\text{sgn}[v_{xT}] = \begin{cases} 1 & \text{if } v_{xT} \geq 0 \\ -1 & \text{if } v_{xT} < 0 \end{cases}$$

The aerodynamic drag force is the result of viscous resistance and pressure distribution over the body of the air working against the motion of the vehicle. The force is given by

$$F_{AD} = \text{sgn}[v_{xT}] \left\{ 0.5 \rho C_D A_F (v_{xT} + v_0)^2 \right\} \quad (2.4)$$

where ρ is the air density in kg/m^3 , C_D is the aerodynamic drag coefficient (dimensionless, and typically is $0.2 < C_D < 0.4$), A_F is the equivalent frontal area of the vehicle, and v_0 is the head-wind velocity.

2.4 DYNAMICS OF VEHICLE MOTION

Tractive force is the force between the vehicle's tires and the road (and parallel to the road) supplied by the electric motor in an EV and by the combination of electric motor and IC engine in an HEV to overcome the road load. The dynamic equation of motion in the tangential direction is given by

$$k_m m \frac{dv_{xT}}{dt} = F_{TR} - F_{RL} \quad (2.5)$$

where k_m is the rotational inertia coefficient to compensate for the apparent increase in the vehicle's mass due to the onboard rotating mass. The typical values of k_m are between 1.08 and 1.1, and it is dimensionless. The acceleration of the vehicle is dv_{xT}/dt .

Dynamic equations can be represented in the state space format for simulation of an EV or HEV system. The motion described by Equation 2.5 is the fundamental equation required for dynamic simulation of the vehicle system. One of the state variables of the vehicle dynamical system is v_{xT} . The second equation needed for modeling and simulation is the velocity equation, where either s or x_F can be used as the state variable. The slope of the roadway β will be an input to the simulation model, which may be given in terms of the tangential roadway distance as $\beta = \beta(s)$ or in terms of the horizontal distance as $\beta = \beta(x_F)$. If β is given in terms of s , then the second state variable equation is

$$\frac{ds}{dt} = v_{xT} \quad (2.6)$$

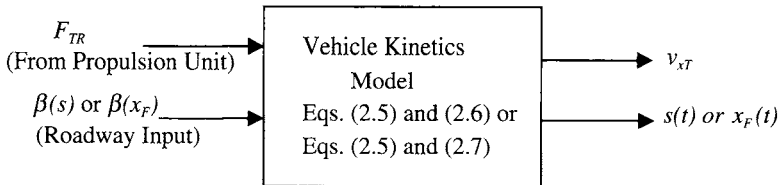


FIGURE 2.9 Modeling of vehicle kinetics and roadway.

If β is given in terms of x_F , then the second state variable equation is

$$\frac{dx_F}{dt} = \frac{v_{xT}}{\sqrt{1 + \left[\frac{df}{dx_F} \right]^2}} \quad (2.7)$$

The input-output relational diagram for simulating vehicle kinetics is shown in Figure 2.9.

2.5 PROPULSION POWER

The desired power rating of the electric motor or the power required from the combination of electric motor and IC engine (i.e., the propulsion unit) can be calculated from the above equations based on the system constraints of starting acceleration, vehicle rated and maximum velocity, and vehicle gradability. The torque at the wheels of the vehicle can be obtained from the power relation:

$$Power = T_{TR} \cdot \omega_{wh} = F_{TR} \cdot v_{xT} \text{ watts} \quad (2.8)$$

where T_{TR} is the tractive torque in $N\cdot m$, and ω_{wh} is the angular velocity of the wheel in rad/s. F_{TR} is in N, and v_{xT} is in m/s. Assuming no slip between the tires and the road, the angular velocity and the vehicle speed are related by

$$v_{xT} = \omega_{wh} \cdot r_{wh} \quad (2.9)$$

where r_{wh} is the radius of the wheel in meters. The losses between the propulsion unit and wheels in the transmission and the differential have to be appropriately accounted for when specifying the power requirement of the propulsion unit.

An advantage of an electrically driven propulsion system is the elimination of multiple gears to match the vehicle speed and the engine speed. The wide speed range operation of electric motors enabled by power electronics control makes it possible to use a single gear-ratio transmission for instantaneous matching of the available motor torque T_{motor} with the desired tractive torque T_{TR} . The gear ratio and the size depend on the maximum motor speed, the maximum vehicle

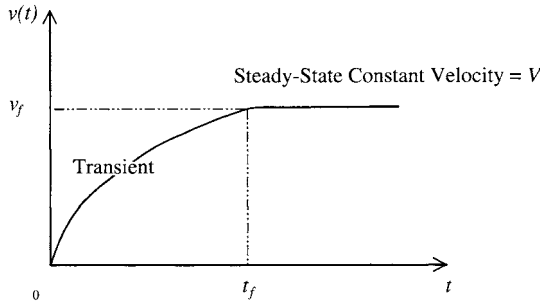


FIGURE 2.10 Plot of velocity profile.

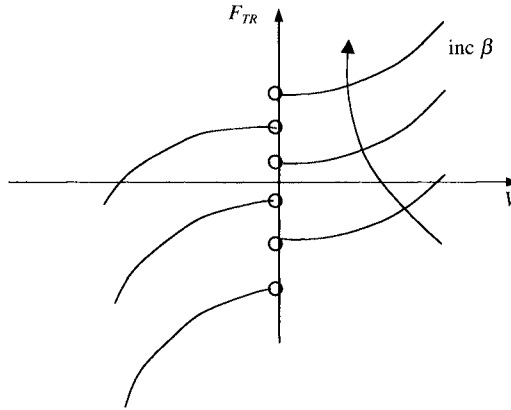


FIGURE 2.11 Tractive force versus steady state velocity characteristics.

speed, the wheel radius, and the available traction between the tires and the road. A higher motor speed relative to the vehicle speed means a higher gear-ratio, larger size, and higher cost. However, higher motor speed is also desired in order to increase the power density of the motor. Therefore, a compromise is necessary between the maximum motor speed and the gear-ratio to optimize the cost. Planetary gears are typically used for EVs, with the gear-ratio rarely exceeding ten.

2.5.1 FORCE-VELOCITY CHARACTERISTICS

Having identified the fundamental forces and the associated dynamics for electric and hybrid electric vehicles, let us now attempt to relate these equations to vehicle design specifications and requirements. For an efficient design of the propulsion unit, the designer must know the force required to accelerate the vehicle to a cruising speed within a certain time and then to propel the vehicle forward at the rated steady state cruising speed and at the maximum speed on a specified slope. Useful design information is contained in the vehicle speed versus time and the steady state tractive force versus constant velocity characteristics, illustrated in Figures 2.10 and 2.11. In the sections to follow and in the remainder of the book, we will always assume the velocity to be in the tangential direction and denote it by v instead of v_{xT} for simplicity. The steady state constant velocity will be denoted by the uppercase letter V .

Tractive force versus steady state velocity characteristics can be obtained from the equation of motion (Equation 2.5). When the steady state velocity is reached, $dv/dt=0$; and, hence, $\Sigma F=0$. Therefore, we have

$$F_{TR} - F_{AD} - F_{ROLL} - F_{g\alpha T} = 0$$

$$\Rightarrow F_{TR} = mg \left[\sin \beta + C_0 \operatorname{sgn}(V) \right] + \operatorname{sgn}(V) \left[mg C_i + \frac{\rho}{2} C_D A_F \right] V^2$$

Note that

$$1. \frac{dF_{TR}}{dV} = 2V \operatorname{sgn}(V) \left(\frac{\rho C_D A_F}{2} + mg C_i \right) > 0 \quad \forall V \quad \text{and}$$

$$2. \lim_{V \rightarrow 0^+} F_{TR} \neq \lim_{V \rightarrow 0^-} F_{TR}$$

The first equation reveals that the slope of F_{TR} versus V is always positive, meaning that the force requirement increases as the vehicle speed increases, which is primarily due to the aerodynamic drag force opposing vehicle motion. Also, discontinuity of the curves at zero velocity is due to the rolling resistance force.

2.5.2 MAXIMUM GRADABILITY

The maximum grade that a vehicle will be able to overcome with the maximum force available from the propulsion unit is an important design criterion as well as performance measure. The vehicle is expected to move forward very slowly when climbing a steep slope and, hence, we can make the following assumptions for maximum gradability:

1. The vehicle moves very slowly $\Rightarrow v \cong 0$.
2. F_{AD} and F_{roll} are negligible.
3. The vehicle is not accelerating, i.e., $dv/dt=0$.
4. F_{TR} is the maximum tractive force delivered by a motor (or motors) at near zero speed.

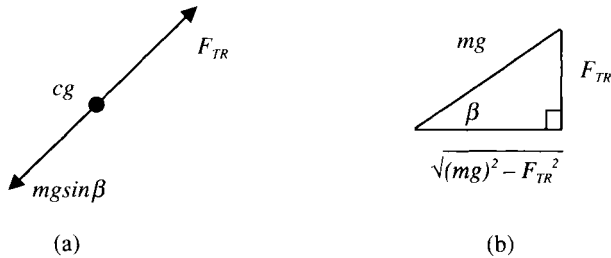


FIGURE 2.12 (a) Force diagram to determine maximum gradability. (b) Tractive force and mg with respect to the grade.

At near stall conditions, under the above assumptions,

$$\sum F = 0 \Rightarrow F_{TR} - F_{gxT} = 0 \Rightarrow F_{TR} = mg \sin \beta$$

Therefore,

$$\sin \beta = \frac{F_{Tr}}{mg}.$$

The maximum percent grade is as follows (Figure 2.12):

$$\begin{aligned} \max \% \text{ grade} &= 100 \tan \beta \\ \Rightarrow \max \% \text{ grade} &= \frac{100 F_{TR}}{\sqrt{(mg)^2 - F_{TR}^2}} \end{aligned} \quad (2.10)$$

2.6

VELOCITY AND ACCELERATION

The energy required from the propulsion unit depends on the desired acceleration and the road load force that the vehicle has to overcome. Maximum acceleration is limited by the maximum tractive power available and the roadway condition at the time of vehicle operation. Although the road load force is unknown in a real-world roadway, significant insight about the vehicle velocity profile and energy requirement can be obtained through studies of assumed scenarios. Vehicles are typically designed with a certain objective, such as maximum acceleration on a given roadway slope in a typical weather condition. Discussed in the following are two simplified scenarios that will set the stage for designing EVs and HEVs.

2.6.1

CONSTANT F_{TR} , LEVEL ROAD

In the first case, we will assume a level road condition, where the propulsion unit for an EV exerts a constant tractive force. The level road condition implies that $\beta(s) = 0$. We will assume that the EV is initially at rest, which implies $v(0) = 0$. The free body diagram at $t=0$ is shown in Figure 2.13.

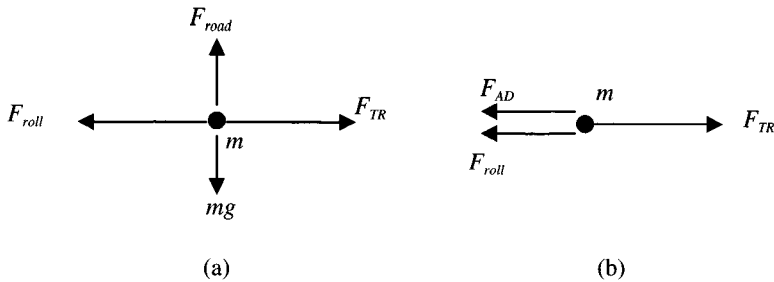


FIGURE 2.13 Forces acting on the vehicle on a level road. (a) Free body diagram $t=0$. (b) Forces on the vehicle at $t > 0$.

Assume that $F_{TR}(0)=F_{TR}>C_0mg$, i.e., the initial tractive force is capable of overcoming the initial rolling resistance. Therefore,

$$\Sigma F(0) = ma(0) = m \frac{dv(0)}{dt}$$

$$\Rightarrow F_{TR} - C_0mg = m \frac{dv(0)}{dt}$$

Because $F_{TR}(0)>C_0mg$, $dv(0)/dt>0$, and the velocity v increases. Once the vehicle starts to move, the forces acting on it change. At $t>0$:

$$\Sigma F = m \frac{dv}{dt} \Rightarrow F_{TR} - F_{AD} - F_{roll} = m \frac{dv}{dt}$$

$$\Rightarrow F_{TR} - \text{sgn}[v(t)] \frac{\rho}{2} C_D A_F v^2(t) - \text{sgn}[v(t)] mg (C_0 + C_1 v^2(t)) = m \frac{dv}{dt}$$

Assuming $v(t)>0$ for $t>0$ and solving for dv/dt ,

$$\frac{dv}{dt} = \left(\frac{F_{TR}}{m} - gC_0 \right) - \left[\frac{\rho}{2m} C_D A_F + gC_1 \right] v^2$$

Let us define the following constants for a constant F_{TR} acceleration:

$$K_1 = \frac{F_{TR}}{m} - gC_0 > 0$$

$$K_2 = \frac{\rho}{2m} C_D A_F + gC_1 > 0$$

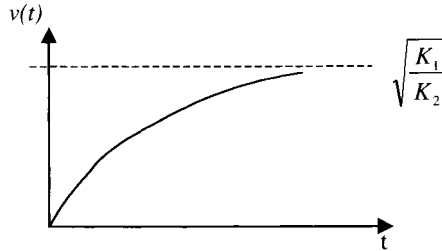


FIGURE 2.14 Velocity profile for a constant F_{tr} on a level road.

which gives

$$\frac{dv}{dt} = K_1 - K_2 v^2$$

2.6.1.1 Velocity Profile

The velocity profile for the constant F_{TR} level road case (Figure 2.14) can be obtained by solving for v from the dv/dt equation above, which gives

$$v(t) = \sqrt{\frac{K_1}{K_2}} \tanh\left(\sqrt{K_1 K_2} t\right) \quad (2.11)$$

The terminal velocity can be obtained by taking the limit of the velocity profile as time approaches infinity. The terminal velocity is

$$V_T = \lim_{t \rightarrow \infty} v(t) = \sqrt{\frac{K_1}{K_2}} \Rightarrow \sqrt{K_1 K_2} = K_2 V_T$$

2.6.1.2 Distance Traversed

The distance traversed by the vehicle can be obtained from the following relation:

$$\frac{ds(t)}{dt} = v(t) = V_T \tanh(K_2 V_T t)$$

The distance as a function of time is obtained by integrating the above equation:

$$s(t) = \frac{1}{K_2} \ln \left[\cosh(K_2 V_T t) \right] \quad (2.12)$$

The starting acceleration is often specified as 0 to v_f m/s in t_f s, where v_f is the desired velocity at the end of the specified time t_f s. The time to reach the desired velocity is given by

$$t_f = \frac{1}{K_2 V_T} \cosh^{-1} \left[e^{(K_2 s_f)} \right] \quad (2.13)$$

and the distance traversed during the time to reach the desired velocity is given by

$$s_f = \frac{1}{K_2} \ln \left[\cosh(K_2 V_T t_f) \right] \quad (2.14)$$

The desired time can also be expressed as follows:

$$t_f = \frac{1}{\sqrt{K_1 K_2}} \tanh^{-1} \left(\sqrt{\frac{K_2}{K_1}} v_f \right) \quad (2.15)$$

where v_f the velocity after time t_f .

Let t_{v_T} = time to reach 98% of the terminal velocity V_T . Therefore,

$$t_{v_T} = \frac{1}{\sqrt{K_1 K_2}} \tanh^{-1} \left(\frac{0.98 V_T}{V_T} \right) \text{ or} \quad (2.16)$$

$$t_{v_T} = \frac{2.3}{\sqrt{K_1 K_2}} \quad \text{or} \quad \frac{2.3}{K_2 V_T}$$

2.6.1.3 Tractive Power

The instantaneous tractive power delivered by the propulsion unit is

$$P_{TR}(t) = F_{TR} v(t)$$

Substituting for $v(t)$,

$$P_{TR}(t) = F_{TR} V_T \tanh(K_2 V_T t)$$

$$\Rightarrow P_{TR}(t) = F_{TR} V_T \tanh(\sqrt{K_1 K_2} t) = P_T \tanh(\sqrt{K_1 K_2} t) \quad (2.17)$$

Terminal power can be expressed as $P_T = F_{TR} V_T$. The tractive power required to reach the desired velocity v_f over the acceleration interval Δt (refer to Figure 2.15) is

$$P_{TR,pk} = P_T \tanh(\sqrt{K_1 K_2} t_f) \quad (2.18)$$

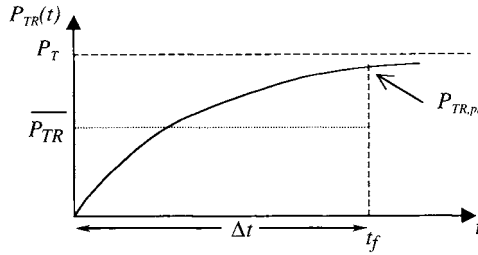


FIGURE 2.15 Acceleration interval $\Delta t = t_f - 0$.

The mean tractive power over the acceleration interval Δt is

$$\overline{P_{TR}} = \frac{1}{t_f} \int_0^{t_f} P_{TR}(t) dt$$

$$\Rightarrow \overline{P_{TR}} = \frac{P_T}{t_f} \frac{1}{\sqrt{K_1 K_2}} \ln \left[\cosh(\sqrt{K_1 K_2} t_f) \right] \quad (2.19)$$

2.6.1.4 Energy Required

The energy requirement for a given acceleration and constant steady state velocity is necessary for the design and selection of the energy source or batteries to cover a certain distance. The rate of change of energy is the tractive power, given as

$$P_{TR}(t) = \frac{de_{TR}}{dt}$$

where e_{TR} is the instantaneous tractive energy. The energy required or energy change during an interval of the vehicle can be obtained from the integration of the instantaneous power equation as

$$\begin{aligned} \int_{e_{TR}(0)}^{e_{TR}(t_f)} de_{TR} &= \int_{t=0}^{t_f} P_{TR} dt \\ \Rightarrow \Delta e_{TR} &= t_f \overline{P_{TR}} \end{aligned} \quad (2.20)$$

EXAMPLE 2.1

An electric vehicle has the following parameter values:

$$m=800 \text{ kg}, C_D=0.2, A_F=2.2 \text{ m}^2, C_0=0.008, C_1=1.6 \times 10^{-6} \text{ s}^2/\text{m}^2$$

Also, take density of air $\rho=1.18 \text{ kg/m}^3$, and acceleration due to gravity $g=9.81 \text{ m/s}^2$.

The vehicle is on level road. It accelerates from 0 to 65 mph in 10 s, such that its velocity profile is given by

$$v(t) = 0.29055t^2 \quad \text{for} \quad 0 \leq t \leq 10 \text{ s}$$

- Calculate $F_{TR}(t)$ for $0 \leq t \leq 10 \text{ s}$.
- Calculate $P_{TR}(t)$ for $0 \leq t \leq 10 \text{ s}$.
- Calculate the energy loss due to nonconservative forces E_{loss} .
- Calculate Δe_{TR} .

Solution

- From the force balance equation,

$$\begin{aligned} F_{TR} - F_{AD} - F_{roll} &= m \frac{dv}{dt} \\ \Rightarrow F_{TR}(t) &= m \frac{dv}{dt} + \frac{\rho}{2} C_D A_F v^2 + mg(C_0 + C_1 v^2) \\ &= 464.88t + 0.2192t^4 + 62.78 \text{ N} \end{aligned}$$

(b) The instantaneous power is

$$\begin{aligned} P_{TR}(t) &= F_{TR}(t) * v(t) \\ &= 135.07t^3 + .00637t^6 + 18.24t^2W \end{aligned}$$

(c) The energy lost due to nonconservative forces is

$$\begin{aligned} E_{loss} &= \int_0^{10} v(F_{AD} + F_{roll})dt = \int_0^{10} 0.29055t^2(0.0219t^4 + 62.78)dt \\ &= 15,180 \text{ J} \end{aligned}$$

(d) The kinetic energy of the vehicle is

$$\Delta KE = \frac{1}{2} m [v(10)^2 - v(0)^2] = 337,677 \text{ J}$$

Therefore, the change in tractive energy is

$$\begin{aligned} \Delta e_{TR} &= 15,180 + 337,677 \\ &= 352,857 \text{ J} \end{aligned}$$

EXERCISE 2.2

An EV has the following parameter values: $\rho=1.16 \text{ kg/m}^3$, $m=692 \text{ kg}$, $C_D=0.2$, $A_F=2 \text{ m}^2$, $g=9.81 \text{ m/s}^2$, $C_\theta=0.009$, and $C_l=1.75*10^{-6} \text{ s}^2/\text{m}^2$. The EV undergoes constant F_{TR} acceleration on a level road starting from rest at $t=0$. The maximum continuous F_{TR} that the electric motor is capable of delivering to the wheels is 450 N.

(a) Find $V_T(F_{TR})$, and plot it.

(b) If $F_{TR}=350\text{N}$:

(i) Find V_T .

(ii) Plot $v(t)$ for $t \geq 0$.

(iii) Find t_{VT}

(iv) Calculate the time required to accelerate from 0 to 60 mph.

(v) Calculate $P_{TR,pk}$, $\overline{P_{TR}}$, and Δe_{TR} corresponding to acceleration to $0.98 V_T$.

Solution

$$(a) V_T(F_{TR}) = 53.2 \sqrt{1.45 \times 10^{-3} F_{TR} - 0.0883} \text{ m/s}$$

(b)

(i) 34.4 m/s

(ii) $v(t)=34.4 \tanh(1.22 \times 10^{-2}t) \text{ m/s}$

(iii) 189 s

(iv) 85.6 s

(v) $P_{TR,pk}=11.8 \text{ kW}$, $\overline{P_{TR}}=8.46 \text{ kW}$, and $\Delta e_{TR}=1.61 \text{ MJ}$

2.6.2 NONCONSTANT F_{TR} , GENERAL ACCELERATION

In the general case, with a nonconstant F_{TR} and an arbitrary velocity profile, as shown in Figure 2.16, force can be calculated as follows:

$$\begin{aligned}
 \Sigma F &= m \frac{dv}{dt} \\
 \Rightarrow F_{TR} - F_{AD} - F_{roll} - F_{gxT} &= m \frac{dv}{dt} \\
 \Rightarrow F_{TR} &= m \frac{dv}{dt} + mg \sin \beta + F_{AD} + F_{roll} \\
 &= m \frac{dv}{dt} + mg \sin \beta + \left[mg C_1 + \frac{\rho}{2} A_F C_D \right] v^2 + mg C_0
 \end{aligned} \tag{2.21}$$

The instantaneous tractive power $P_{TR}(t)$ is

$$\begin{aligned}
 P_{TR}(t) &= F_{TR}(t) v(t) \\
 &= mv \frac{dv}{dt} + v (F_{gxT} + F_{AD} + F_{roll})
 \end{aligned} \tag{2.22}$$

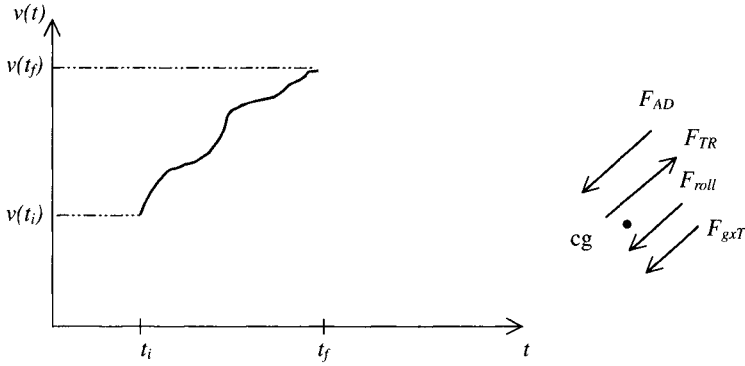


FIGURE 2.16 Arbitrary velocity profile.

The change in tractive energy Δe_{TR} is

$$\begin{aligned}
 \Delta e_{TR} &= \int_{t_i}^{t_f} P_{TR}(t) dt \\
 &= m \int_{v(t_i)}^{v(t_f)} v dv + \int_{t_i}^{t_f} (v) F_{gxT} dt + \int_{t_i}^{t_f} (v) (F_{AD} + F_{roll}) dt
 \end{aligned} \tag{2.23}$$

The energy supplied by the propulsion unit is converted into various forms of energy, some of which are stored in the vehicle system, while others are lost due to non-constructive forces. It is interesting to note the type of energy associated with each term in Equation 2.23. Let us consider the first term on the right side of Equation 2.23:

$$m \int_{v(t_i)}^{v(t_f)} v dv = \frac{1}{2} m \left[v^2(t_f) - v^2(t_i) \right] = \Delta(\text{Kinetic Energy})$$

Also,

$$\begin{aligned} \int_{t_i}^{t_f} (v) F_{gxT} dt &= mg \int_{t_i}^{t_f} v \sin \beta dt = mg \int_{s(t_i)}^{s(t_f)} \sin \beta ds = mg \int_{f(t_i)}^{f(t_f)} df \\ &= mg \left[f(t_f) - f(t_i) \right] \\ &= \Delta(\text{Potential Energy}) \end{aligned}$$

The above term represents change in vertical displacement multiplied by mg , which is the change in potential energy.

The third and fourth terms on the right side of Equation 2.23 represent the energy required to overcome nonconstructive forces that include rolling resistance and aerodynamic drag force. The energy represented in these terms is essentially the loss term. Therefore,

$$\int_{t_i}^{t_f} (v) (F_{AD} + F_{roll}) dt = E_{loss}$$

Let

$$K_3 = mg C_0, \quad K_4 = mg C_1 + \frac{\rho}{2} C_D A_F$$

For

$$v(t) > 0, \quad t_i \leq t \leq t_f,$$

$$\begin{aligned} E_{loss} &= K_3 \int_{t_i}^{t_f} \frac{ds}{dt} dt + K_4 \int_{t_i}^{t_f} v^3 dt; \\ &= K_3 \Delta s + K_4 \int_{t_i}^{t_f} v^3 dt \end{aligned}$$

In summary, we can write

$$\Delta e_{TR} = \frac{1}{2} m \left[v^2(t_f) - v^2(t_i) \right] + mg \left[f(t_f) - f(t_i) \right] + \int_{t_i}^{t_f} (v)(F_{AD} + F_{roll}) dt$$

or

$$\Delta e_{TR} = \Delta(\text{Kinetic Energy}) + \Delta(\text{Potential Energy}) + E_{loss}$$

EXERCISE 2.3

The vehicle with parameters as given in Exercise 2.2 accelerates from 0 to 60 mi/h in 13.0 s for the two following acceleration types: constant F_{TR} and uniform acceleration.

- Plot, on the same graph, the velocity profile of each acceleration type.
- Calculate and compare the tractive energy required for the two types of acceleration. $F_{TR} = \text{const.} = 1548 \text{ N}$

Solution

(a) $\Delta e_{TR} = 0.2752 \text{ MJ}$ for constant F_{TR} , and $\Delta e_{TR} = 0.2744 \text{ MJ}$ for uniform acceleration.

2.7

PROPULSION SYSTEM DESIGN

The steady state maximum velocity, maximum gradability, and velocity equations can be used in the design stage to specify the power requirement of a particular vehicle. The common design requirements related to vehicle power expected to be specified by a customer are the initial acceleration, rated velocity on a given slope, maximum % grade, and maximum steady state velocity.

The complete design is a complex issue involving numerous variables, constraints, considerations, and judgment, which is beyond the scope of this book. The basic equations and principles that lay the foundation for design of EVs and HEVs are presented in this book. Design considerations will be highlighted as and when appropriate. Let us consider the tractive power requirement for initial acceleration, which plays a significant role in determining the rated power of the propulsion unit. The initial acceleration is specified as 0 to v_f in t_f s. The design problem is to solve for F_{TR} , starting with a set of variables including vehicle mass, rolling resistance, aerodynamic drag coefficient, percent grade, wheel radius, etc., some of which are known, while others have to be assumed. The acceleration of the vehicle in terms of these variables is given by

$$a = \frac{dv}{dt} = \frac{F_{TR} - F_{RL}}{m} \quad (2.24)$$

The tractive force output of the electric motor for an EV or the combination of electric motor and IC engine for an HEV will be a function of vehicle velocity. Furthermore, the road load characteristics are a function of velocity, resulting in a transcendental equation to be solved to determine the desired tractive power from the propulsion unit. Other design requirements also play a significant role in determining tractive power. The problem is best handled by a computer simulation, with which the various equations of Sections 2.5, 2.6, and 2.7 can be used iteratively to calculate the tractive force and power requirements from the propulsion unit for the given set of specifications. Insight gained from the scenarios discussed in this chapter will be used later to specify the power ratings of the electric motor for EVs and HEVs in Chapter 9 and for IC engines for HEVs in Chapter 10.

PROBLEMS

2.1

A length of road is straight, and it has a profile in the x - y plane described by

$$f(x) = 200 \ln[7.06 * 10^{-4} 9x + 1416]$$

where $0 \leq x \leq 3$ miles = 15,840 ft; $f(x)$ and x are in feet.

- Plot the road profile in the x - y plane for $0 \leq x \leq 15,840$ ft.
- Derive an expression for $\beta(x)$. Calculate $\beta(500$ ft).
- Derive an expression for percent grade (x). Calculate percent grade (500 ft).
- Derive an expression for tangential road length $s(x)$, such that $s(0)=0$. Calculate $s(500$ ft).
- Can you find an expression for $x(s)$, i.e., can you express x as a function of s ? Show some steps in your attempt.

2.2

A straight roadway has a profile in the x - y plane given by

$$f(x_f) = 4.1\sqrt{x_f} \quad \text{for } 0 \leq x_f \leq 2 \text{ mile} = 10,560 \text{ ft}$$

$f(x_f)$ and x_f are measured in feet.

- Derive an expression for $\beta(x_f)$. Calculate $\beta(1$ mi).
- Calculate the tangential road length, s from 0 to 2 mi.

2.3

An electric vehicle has the following parameter values:

$$m = 692 \text{ kg}, C_D = 0.2, A_F = 2 \text{ m}^2, C_0 = 0.009, \text{ and } C_1 = 1.75 * 10^{-6} \text{ s}^2/\text{m}^2$$

Also, take $\rho = 1.16 \text{ kg/m}^3$ and $g = 9.81 \text{ m/s}^2$.

- (a) *EV at rest*—The EV is stopped at a stop sign at a point in the road where the grade is +15%. The tractive force of the vehicle is supplied by the vehicle brakes.
- Calculate the tractive force necessary for zero rolling resistance. (The vehicle is at rest.)
 - Calculate the minimum tractive force required from the brakes to keep the EV from rolling down the grade.
- (b) *EV at constant velocity*—The EV is moving at a constant velocity along a road that has a constant grade of -12%.
- Plot, on the same graph, the magnitudes of the tangential gravitational force (F_{gxT}), the aerodynamic drag force (F_{AD}), and the rolling resistance force (F_{roll}) versus velocity for $0 < V \leq 180$ mph. Over that range of velocity, does F_{gxT} dominate? When does F_{AD} dominate? When does F_{roll} dominate? Label these regions on the graph.
 - Derive an expression for the tractive force as a function of velocity. Plot this expression on its own graph. Is the tractive force always in the same direction?

2.4

Showing all steps, derive and plot the velocity profile [i.e., $v(t)$] for constant F_{TR} -constant grade acceleration. (Constant grade means that β is constant but not necessarily zero.) Given:

- The EV starts from rest at $t=0$.
- The resultant of F_{gxT} and F_{TR} is enough to overcome the rolling resistance to get the EV moving.
- $v(t) \geq 0$ for $t \geq 0$.

What effect does gravity have on the velocity profile?

2.5

A vehicle accelerates from 0 to 60 mph in 10 s with the velocity profile given by

$$v(t) = 20 \ln(0.282t + 1) \text{ m/s} \quad \text{for} \quad 0 \leq t \leq 10 \text{ s}$$

The vehicle is on level road. For the problem, use the parameters given in Problem 2.3.

- Calculate and plot $F_{TR}(t)$ and $P_{TR}(t)$ for $0 \leq t \leq 10$ s
- Calculate Δe_{TR} . How much of Δe_{TR} is ΔKE ? How much is E_{loss} ?

2.6

Using the vehicle parameters given in Problem 2.3, calculate and plot, on the same graph, steady state F_{TR} versus V characteristics for $\beta = \pm 4^\circ$ and $-60 \text{ mph} \leq V \leq 60 \text{ mph}$, but $V \neq 0$.

2.7

An electric vehicle racer will attempt to jump five city buses as shown in Figure P2.7. The vehicle will start at rest at a position 100 m from the beginning of the take-off ramp. The vehicle will accelerate uniformly, until it reaches the end of the take-off ramp, at which time it will be traveling at 100 mph.

The vehicle has the following parameter values:

$$m = 692 \text{ kg}, C_D = 0.2, A_f = 2 \text{ m}^2, C_0 = 0.009, \text{ and } C_1 = 1.75 * 10^{-6} \text{ s}^2/\text{m}^2$$

Also, take density of air $\rho = 1.16 \text{ kg/m}^3$, and acceleration due to gravity $g = 9.81 \text{ m/s}^2$.

- Sketch and label the velocity profile of the vehicle from the time it starts to the time it reaches the end of the take-off ramp. How much time does the vehicle take to reach the end of the take-off ramp?
- Calculate the change in gravitational potential energy, from the start to the end of the take-off ramp.
- Calculate energy loss, E_{loss} , from the start to the end of the take-off ramp, if $\Delta e_{TR} = 8.28 \times 10^5 \text{ J}$ during that period.

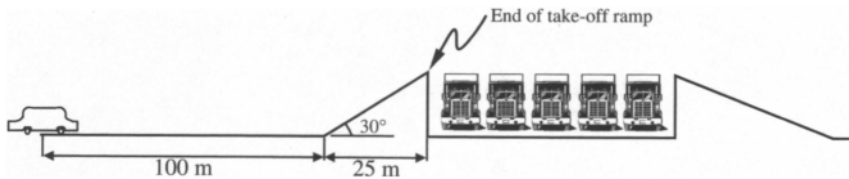


FIGURE P2.7

reference frame transformations, but today's digital processors are capable of completing complex algorithms in a relatively short time.

The competitor to the induction motor is the permanent magnet (PM) motor. The permanent magnet AC motors have magnets on the rotor, while the stator construction is the same as that of the induction motor. The PM motors can be of surface-mounted type, or the magnets can be inset within the rotor in the interior PM motors. The PM motor can also be classified as sinusoidal type or trapezoidal type depending on the flux distribution in the air gap. Trapezoidal motors have concentrated three-phase windings and are also known as brushless DC motors. The PM motors are driven by a six-switch inverter just like an induction motor, but the control is relatively simpler than that of the induction motor. The use of high-density rare earth magnets in PM motors provides high power density, but at the same time, the cost of magnets is on the negative side for these motors. For EV and HEV applications, motor size is relatively large compared to the other smaller power applications of PM motors, which amplifies the cost problem. However, HEV motors are usually smaller than EV motors, and the performance and efficiency achievable from PM motors may be enough to overcome the cost problem. Interior PM motors have excellent performance characteristics, superior to surface mount magnets, but manufacturing complexity is one of the drawbacks for these motors.

Another candidate for traction motors is the switched reluctance (SR) motor. These motors have excellent fault tolerance characteristics, and their construction is fairly simple. The SR motors have no windings, magnets, or cages on the rotor, which helps increase the torque and inertia rating of these motors. The motor speed-torque characteristics are an excellent match with the road load characteristics, and performance of SR motors for EV/HEV applications have been found to be excellent. The two problems associated with SR motors are acoustic noise and torque ripple. The noise problems of SR motors are being addressed in various research activities.

5.1 MOTOR AND ENGINE RATINGS

The strength of electric motors and IC engines are typically described with "hp" or horsepower ratings, although a comparison between electric motors and IC engines in terms of hp only is not fair. The power that an electric motor can continuously deliver without overheating is its rated hp, which is typically a derated figure. For short periods of time, the motor can deliver two to three times the rated hp. Therefore, at starting, high power is available from an electric motor for acceleration, and the motor torque can be maximum under stall conditions, i.e., at zero speed. Motor type determines whether maximum torque is available at zero speed or not. On the contrary, an IC engine is rated at a specific r/min level for maximum torque and maximum hp. The IC engine maximum torque and hp ratings are typically derived under idealized laboratory

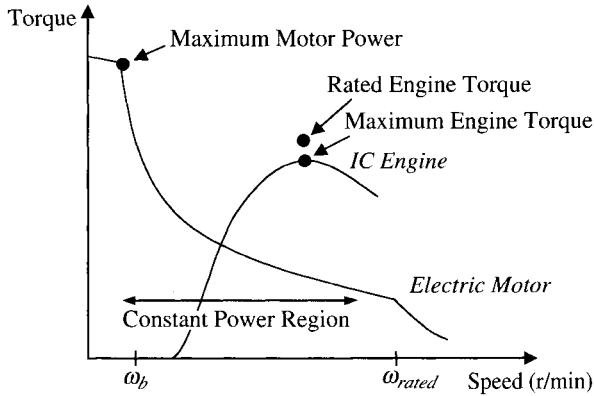


FIGURE 5.1 Electric motor and IC engine torque characteristics.

conditions. In practical situations, it is impossible to achieve the rated hp; the maximum hp available from an IC engine is always smaller than the rated hp.

The torque characteristics of motors are shown in [Figure 5.1](#), along with torque characteristics of IC engines. The characteristics of specific motors and IC engines will differ somewhat from these generalized curves. For electric motors, a high torque is available at starting, which is the peak torque of the motor. The peak torque is much higher (typically twice) than that of the rated torque. The peak torque for electric motors in an EV/HEV application needs to be sustained for about 60 to 90 s. The *rated torque* for electric motors is the torque that must be available for an indefinite length of time. The peak or rated power from a motor is obtained at base speed (ω_b) when motor characteristics enter the constant power region from the constant torque region, once the voltage limit of the power supply is reached. The motor rated speed (ω_{rated}) is at the end of the constant power region. The IC engine peak power and torque occur at the same speed. At this stage, it is helpful to review the power and torque relation, which is as follows:

$$\text{Power (Watts)} = \text{Torque (N-m)} \times \text{speed (rads/s)} \quad (5.1)$$

The power-torque relation in hp and ft-lb is

$$HP = \frac{\text{Torque (ft-lbs)} \times \text{rpm}}{5232}$$

[Figure 5.1](#) depicts that the IC engine does not produce torque below a certain speed. A transmission is essential for an IC engine to match the vehicle speed with the narrow high hp speed range of the engine. On the other hand, the electric motor produces high torque even at zero speed and typically has constant power characteristics over a wide speed range. Therefore, the electric motor can be attached directly to the drive wheels and accelerate the vehicle from zero speed all the way to top speed. The motor and, hence, the vehicle speed can be

controlled directly through the power electronic converter feeding the current into the motor. There is essentially no requirement for a transmission with an electric motor, other than a fixed gear for appropriately sizing the motor.

5.2

EV AND HEV MOTOR REQUIREMENTS

The important characteristics of a motor for an EV or HEV include flexible drive control, fault tolerance, high efficiency, and low acoustic noise. The motor drive must be capable of handling voltage fluctuations from the source. Another important requirement of the electric motor is acceptable mass production costs, which is to be achieved through technological advancement. The requirements of an EV or HEV motor, not necessarily in order of importance, are itemized in the following:

- Ruggedness
- High torque-to-inertia ratio (T_e/J); large T_e/J results in “good” acceleration capabilities
- Peak torque capability of about 200 to 300% of continuous torque rating
- High power-to-weight ratio (P_e/w)
- High-speed operation, ease of control
- Low acoustic noise, low electromagnetic interference (EMI), low maintenance, and low cost
- Extended constant power region of operation

5.3

DC MACHINES

The torque in electric machines is produced utilizing one of two basic principles of electromagnetic theory: by Lorentz force principle, where torque is produced by the mutual interaction of two orthogonal magnetomotive forces (mmf); and by reluctance principle, where the rotor produces torque while moving toward the minimum reluctance position in a varying reluctance path. The DC and AC machines, including the permanent magnet machines, work on the first principle, while the switched reluctance machines work on the latter principle.

The DC machines have two sets of windings, one in the rotor and the other in the stator, which establish the two fluxes; hence, the mmfs that interact with each other produce the torque. The orthogonality of the two mmfs, which is essential for maximum torque production, is maintained by a set of mechanical components called commutators and brushes. The winding in the rotor is called the armature winding, while the winding in the stationary part of the machine is called the field winding. The armature and the field windings are supplied with DC currents. The armature windings carry the bulk of the current, while the field windings carry a small field excitation current. The armature and the field

10

Hybrid Electric Vehicles

A hybrid electric vehicle (HEV) is a vehicle in which at least one of the energy sources, stores, or converters can deliver electric energy. A hybrid road vehicle is one in which the propulsion energy during specified operational missions is available from two or more kinds or types of energy stores, sources, or converters, of which at least one store or converter must be on board. The second definition of hybrid road vehicle is proposed by Technical Committee 69 of Electric Road Vehicles of the International Electrotechnical Commission.

The HEV serves as a compromise for the environmental pollution problem and the limited range capability of today's purely electric vehicle. HEVs have an electric motor as well as an internal combustion engine (ICE) to provide extended range and to "curve down" the pollution problem. Vehicle design complexity increases significantly with hybrid vehicles, because controls and support systems are needed for a thermal engine and an electric machine in addition to the components needed for controlled blending of power coming from the two sources. The hybrids are looked upon by many as a short-term solution until the range limitation and infra-structure problems of purely electric vehicles are solved. Nevertheless, a number of automotive manufacturers are marketing hybrid vehicles for the general population, with many others following suit.

This chapter focuses on familiarizing the reader with the basic drivetrains of HEVs. The basic thermodynamics of some of the heat engines or ICEs are discussed prior to discussing the design issues with HEVs.

10.1

TYPES OF HYBRIDS

10.1.1

SERIES AND PARALLEL HEVs

HEVs evolved out of two basic configurations: series and parallel. A series hybrid is one in which only one energy converter can provide propulsion power. The heat engine or ICE acts as a prime mover in this configuration to drive an electric generator that delivers power to the battery or energy storage link and the

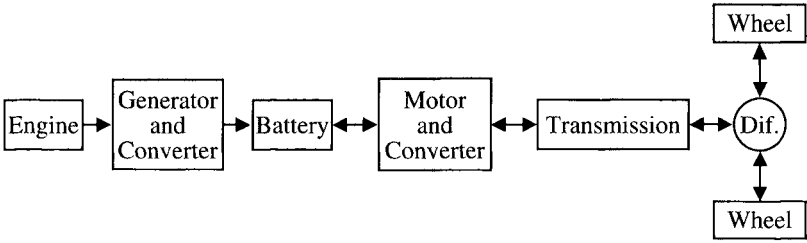


FIGURE 10.1 Series HEV drivetrain.

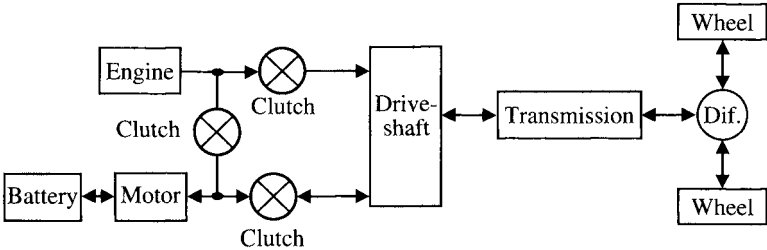


FIGURE 10.2 Parallel HEV drivetrain.

propulsion motor. The component arrangement of a series HEV is shown in [Figure 10.1](#).

A parallel hybrid is one in which more than one energy source can provide propulsion power. The heat engine and the electric motor are configured in parallel, with a mechanical coupling that blends the torque coming from the two sources. The component arrangements of a parallel hybrid are shown in [Figure 10.2](#).

Series HEV is the simpler type, where only the electric motor provides all the propulsion power. A downsized heat engine on board drives a generator, which supplements the batteries and can charge them when they fall below a certain state of charge. The power required to move the vehicle is provided solely by the electric motor. Beyond the heat engine and the generator, the propulsion system is the same as in an EV, making electric motor power requirements the same as for in the EV.

In parallel HEV, the heat engine and the electric motor are connected to the driveshaft through separate clutches. Power requirements of the electric motor in the parallel hybrid are lower than that of an EV or series hybrid, because the heat engine complements for the total power requirement of the vehicle. The propulsion power may be supplied by the heat engine, by the battery-motor set, or by the two systems in combination.

Series and parallel hybrids come in a variety of types. The mission of the vehicle and the optimum design for that mission dictate the choice. If the HEV is to be basically an EV with an ICE-assist for achieving acceptable range, then the

choice should be a series hybrid, with the ICE ensuring that the batteries remain charged all the time. On the other hand, if the HEV is to be basically a vehicle with almost all the performance characteristics and comforts of an ICEV but with lower emission and fuel usage standards, then the choice should be a parallel configuration. Parallel HEVs have been built with performance that is equal, in all aspects of normal operation, to that of a conventional car. However, some series HEVs have also been built that perform nearly as well as ICEVs.

10.1.1.1

Advantages and Disadvantages

The advantages and disadvantages of series and parallel hybrids are summarized in the following. The advantages of a series HEV are:

1. Flexibility of location of engine-generator set
2. Simplicity of drivetrain
3. Suitability for short trips

The disadvantages of a series HEV are:

1. It needs three propulsion components: ICE, generator, and motor.
2. The motor must be designed for the maximum sustained power that the vehicle may require, such as when climbing a high grade. However, the vehicle operates below the maximum power most of the time.
3. All three drivetrain components need to be sized for maximum power for long-distance, sustained, high-speed driving. This is required, because the batteries will exhaust fairly quickly, leaving ICE to supply all the power through the generator.

The following are advantages of a parallel HEV:

1. It needs only two propulsion components: ICE and motor/generator. In parallel HEV, the motor can be used as the generator and vice versa.
2. A smaller engine and a smaller motor can be used to get the same performance, until batteries are depleted. For short-trip missions, both can be rated at half the maximum power to provide the total power, assuming that the batteries are never depleted. For long-distance trips, the engine may be rated for the maximum power, while the motor/generator may still be rated to half the maximum power or even smaller.

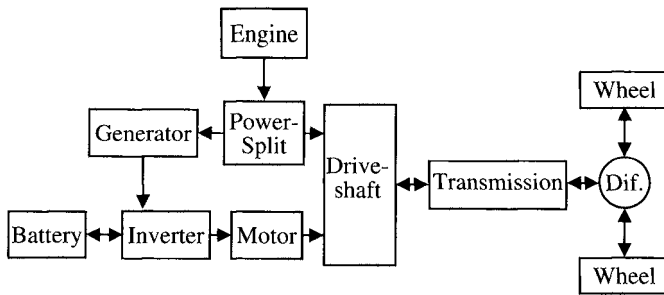


FIGURE 10.3 Series-parallel combination HEV.

The following are disadvantages of a parallel HEV:

1. The control complexity increases significantly, because power flow has to be regulated and blended from two parallel sources.
2. The power blending from the ICE and the motor necessitates a complex mechanical device.

10.1.2

SERIES-PARALLEL COMBINATION

Although HEVs initially evolved as series or parallel, manufacturers later realized the advantages of a combination of the series and parallel configurations for practical road vehicles. In these combination hybrids, the heat engine is also used to charge the battery. The recently available Toyota Prius is an example of such a hybrid, where a small series element is added to the primarily parallel HEV. The small series element ensures that the battery remains charged in prolonged wait periods, such as at traffic lights or in a traffic jam. These combination hybrids can be categorically classified under parallel hybrids, because they retain the parallel structure of a component arrangement. It is important to stress the fact that the detailed configuration of an HEV depends on the application and the trade-off between cost and performance.

The component arrangement of a series-parallel combination hybrid is shown in Figure 10.3. The schematic is based on the Toyota Prius hybrid design. A power split device allocates power from the ICE to the front wheels through the driveshaft and the electric generator, depending on the driving condition. The power through the generator is used to charge the batteries. The electric motor can also deliver power to the front wheels in parallel to the ICE. The inverter is bidirectional and is used to charge the batteries from the generator or to condition the power for the electric motor. For short bursts of speed, power is delivered to the driveshaft from the ICE and the electric motor. A central control unit

regulates the power flow for the system using multiple feedback signals from the various sensors.

Use of the ICE to charge the batteries should be minimized when maximizing efficiency. Energy is always lost while charging and discharging the battery and during the power flow through the inverter. The vehicle should be operated off its engine or battery or both, until the battery is at a minimum acceptable state of charge, say 20 to 40%. The battery should be charged from the power grid when convenient.

10.2 INTERNAL COMBUSTION ENGINES

The devices that convert heat transfer to work cyclically are known as heat engines. Each cycle in the heat engine consists of several different *processes* or *strokes* (such as constant volume, constant pressure, and constant entropy processes) to convert thermal energy into useful work. Different types of cycles are available to design practical heat engines. The practical cycles have evolved due to the limitations of an ideal thermodynamics cycle (such as the Carnot cycle) that operates between the same heat addition and rejection temperatures, and also because of the difference in characteristics of the choices available for the energy source, working fluid, and hardware materials. It must be mentioned that although the heat engines described in the following undergo *mechanical cycles*, the working fluids do not execute a *thermodynamic cycle* because matter is introduced with one composition and is later discharged at a different composition.^{1,2}

The heat engines of interest for EV and HEV applications, primarily the ICE and the gas turbine, will be discussed in this section. An ICE is a heat engine that utilizes fuel as a working fluid. The ICE uses heat cycles that gain its energy from the combustion of fuel within the engine. ICEs can be reciprocating type, where the reciprocating motion of a piston is converted to rotary motion through a crank mechanism. ICEs used in automobiles, trucks, and buses are of the reciprocating type, where the processes occur within a reciprocating piston-cylinder arrangements. The gas turbines used in power plants are also ICEs, where the processes occur in an interconnected series of different components. The Brayton cycle gas turbine engine has been adapted to the automotive propulsion engine and has the advantage of burning fuel that requires little refining and fuel that burns completely. Gas turbines have fewer moving parts, because there is no need to convert the rotary motion of the turbine. The disadvantages of gas turbines are complex construction and lower efficiency. Nevertheless, gas turbines are being considered for HEVs, and prototype vehicles have been developed.

The performance of a heat engine is measured by the efficiency of the heat engine cycle, defined as the ratio of the net work output per cycle W_{net} to the heat transfer into the engine per cycle. Another way of defining the performance of

heat engines is to use the *mean effective pressure* P_{me} . The P_{me} is the theoretical constant gage pressure that, if exerted on the piston during the expansion stroke between the largest specific volume and the smallest volume, would produce the same net work as actually produced by the heat engine. Mathematically stated,

$$P_{me} = \frac{|W_{net}|}{\text{displacement volume}}$$

The heat engine cycle performance analysis is carried out from the information available at certain convenient state points in the cycle. The parameters needed at the state points are pressure, temperature, volume, and entropy. If two parameters are known at two state points, then the unknown parameters are usually obtained from the process that the working fluid undergoes between the two state points (such as constant pressure, isentropic, etc.) and the laws of thermodynamics. Discussions of the laws of thermodynamics and efficiency analysis of heat engine cycles are beyond the scope of this book. Only a general introduction to the heat engine cycles of interest will be given.

Prior to continuing the discussion on ICEs, a few words about entropy are in order. Entropy is a property that is specified for every equilibrium state of a substance. Like energy, entropy is an abstract concept that is extensively used in thermodynamic analysis. Entropy represents the microscopic disorder or uncertainty of a system. Because entropy is a property, the change in entropy in going from one state to another is the same for all processes. The SI unit for specific entropy is J/K.

The reciprocating engines are considered first in the following section to be followed by a discussion on gas turbines.

10.2.1 RECIPROCATING ENGINES

The two types of reciprocating ICEs are the spark-ignition engine (SI) and the compression-ignition (CI) engine. The two engines are commonly known as gasoline/petrol engine and diesel engine, based on the type of fuel used for combustion. The difference in the two engines is in the method of initiating the combustion and in the processes of the cycle. In a SI engine, a mixture of air and fuel is drawn in, and a spark plug ignites the charge. The intake of the engine is called the *charge*. In a CI engine, air is drawn in and compressed to such a high pressure and temperature that combustion starts spontaneously when, later, fuel is injected. The SI engines are relatively light and lower in cost and are used for lower-power engines, as in conventional automobiles. The CI engines are more suitable for power conversion in higher power range, such as in trucks, buses, locomotives, ships, and in auxiliary power units. Their fuel economy is better than the SI engines, justifying their use in higher-power applications.¹

A sketch of a representative reciprocating ICE, including the special terms standard for such engines, is given in [Figure 10.4](#). The engine consists of a piston